

# CCP-CNS2026 — Computer Networks

## Problem 1: Multi-Site Enterprise Network

### GROUP CONTRACT & TEAM CHARTER

|                   |                                                                                                                             |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Instructor        | Sir Muhammad Fahad Irshad                                                                                                   |
| Course            | Computer Network Lab                                                                                                        |
| Problem Assigned  | Problem 1 — Multi-Site Enterprise                                                                                           |
| Team Size         | 4 members                                                                                                                   |
| Project Duration  | 5 Weeks                                                                                                                     |
| Date of Charter   | May 2026                                                                                                                    |
| GitHub Repository | <a href="https://github.com/ZohaibAhmad260/CCP-CNS2026-Problem1">https://github.com/ZohaibAhmad260/CCP-CNS2026-Problem1</a> |
| Section           | V9                                                                                                                          |

## 1. TEAM MEMBERS & ROLES

| # | Name                               | Role                                   | Primary Responsibilities                                                                                                                          |
|---|------------------------------------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Zohaib Ahmad<br>(F2024266251)      | Project Manager &<br>Network Architect | Team lead, topology design, IP addressing plan, VLAN & routing configuration, meeting scheduling, GitHub management, weekly status reports        |
| 2 | M. Efan<br>Amir(F20242661144)      | Security Engineer                      | ACL configuration, DHCP snooping, attack simulation from Kali Linux, DoS execution, post-attack mitigation, incident response                     |
| 3 | Muhammad Abdullah<br>(F2024266337) | Automation Lead &<br>Testing/QA        | GitHub repository setup & maintenance, Python scripts, Wireshark PCAP capture, traffic analysis, ping test matrix, performance testing with iPerf |
| 4 | Tayyab Hassan<br>(F2024266511)     | Documentation<br>Lead                  | Weekly reports, final report compilation (15-20 pages), network diagrams in draw.io, video script, presentation slides, peer evaluation forms     |

## 2. PROJECT OBJECTIVES

This team is assigned Problem 1: Multi-Site Enterprise. The objective is to simulate a secure company network for Apex Manufacturing with the following goals:

- Design and implement a multi-VLAN network (Sales, Engineering, Admin) using GNS3
- Configure a Cisco router using router-on-a-stick for inter-VLAN routing
- Implement DHCP server on router for all three VLANs
- Deploy an Ubuntu web server in the Admin VLAN
- Implement DHCP snooping and ACLs to block Sales from accessing Admin

- Simulate a DoS attack from Kali Linux and capture traffic with Wireshark
- Mitigate the attack using ACL rules and document the full incident

### 3. COMMUNICATION & MEETING PLAN

|                              |                                                 |
|------------------------------|-------------------------------------------------|
| <b>Meeting Frequency</b>     | Minimum twice per week (flexible as needed)     |
| <b>Meeting Platform</b>      | WhatsApp group call / Discord / In-person       |
| <b>Primary Communication</b> | WhatsApp group chat for all members             |
| <b>File Sharing</b>          | GitHub repository (all submissions pushed here) |
| <b>Response Time</b>         | All messages to be acknowledged within 12 hours |

### 4. FIVE-WEEK SCHEDULE OVERVIEW

| Week          | Theme                 | Key Deliverables                                                                          |
|---------------|-----------------------|-------------------------------------------------------------------------------------------|
| <b>Week 1</b> | Planning & Design     | Team charter, topology diagram, IP plan, GitHub setup, risk matrix, resource acquisition  |
| <b>Week 2</b> | GNS3 Implementation   | GNS3 project file, all device configs, ping test matrix, VLAN screenshots, SSH proof      |
| <b>Week 3</b> | Security & Attack     | ACL configs, DHCP snooping, DoS attack script, PCAP files, mitigation proof, attack video |
| <b>Week 4</b> | Integration & Testing | Final GNS3 project, NAT config, performance baseline, full test matrix with pass/fail     |
| <b>Week 5</b> | Final Submission      | Final report (15-20 pages), demo video (8-10 min), presentation slides, peer evaluations  |

### 5. TEAM AGREEMENTS & CODE OF CONDUCT

#### 5.1 Work Ethics

- Every member will complete their assigned role tasks on time
- No member will submit work they did not personally contribute to
- All code, configs, and documents must be pushed to GitHub — no personal storage

#### 5.2 Late Work Policy

- If a member cannot complete their task, they must inform the at least 24 hours before the deadline
- The instructor's policy applies: 10% deduction per day, maximum 50%
- Internally, repeated lateness (2+ times) will be reported to the instructor

## 5.3 GitHub Usage Rules

- All deliverables must be pushed to the team GitHub repository before submission
- Commit messages must be descriptive (e.g., 'Added Week 1 IP addressing plan')
- Muhammad Abdullah (Automation Lead) is the repository admin

## 5.4 Conflict Resolution

- Disagreements are first discussed in the group chat
- Personal conflicts are kept out of project work

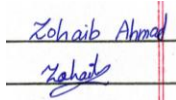
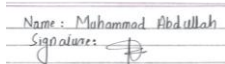
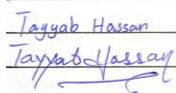
## 6. GRADING AWARENESS

All team members have read and understood the grading structure:

| Component                  | Team Grade Weight         | Points         |
|----------------------------|---------------------------|----------------|
| Week 1 Deliverables        | —                         | 15 pts         |
| Week 2 Deliverables        | —                         | 20 pts         |
| Week 3 Deliverables        | —                         | 25 pts         |
| Week 4 Deliverables        | —                         | 20 pts         |
| Week 5 Deliverables        | —                         | 20 pts         |
| <b>Final Grade Formula</b> | 70% Team + 30% Individual | <b>100 pts</b> |

## 7. SIGNATURES & AGREEMENT

By signing below, each team member confirms that they have read, understood, and agree to all terms in this Group Contract & Team Charter.

| Full Name         | Role                   | Signature                                                                            |
|-------------------|------------------------|--------------------------------------------------------------------------------------|
| Zohaib Ahmad      | PM & Network Architect |  |
| M.Efan Amir       | Security Engineer      |  |
| Muhammad Abdullah | Automation Lead & QA   |  |
| Tayyab Hassan     | Documentation Lead     |  |